

Here are some exam-style ODE problems to prepare you for the style of question you might see on the final exam.

Note: Some solutions use the integrating factor to solve first-order ODEs, because this was historically taught in this course. This year, showing a given solution is correct by differentiating to check will be sufficient. Students are welcome to use the integrating factor approach if they prefer it.

Q1: Consider the ODE

$$\dot{v} = g \cdot (D - v) + B, \quad (1)$$

where g , D , and B are constants. Show that

$$v(t) = \alpha(t) \cdot v_0 + [1 - \alpha(t)] \cdot v_\infty \quad (2)$$

is a solution to ODE (1) with initial conditions $v(t = 0) = v_0$, where

$$\begin{aligned} \alpha(t) &= e^{-gt} \\ v_\infty &= D + B/g \end{aligned} \quad (3)$$

(Any approach permitted).

Approach 1: Verify

This is meant to be a faster approach for students less-practices in solving ODEs directly, but this is not the only or best method — any other approach is fine. We are given

$$\begin{aligned} v(t) &= \alpha(t) \cdot v_0 + [1 - \alpha(t)] \cdot v_\infty \\ &= v_\infty + e^{-gt}(v_0 - v_\infty); \end{aligned} \quad (4)$$

Treat this as an ansatz and differentiate to verify:

$$\begin{aligned} \dot{v} &= -ge^{-gt}(v_0 - v_\infty) \\ &= -g(v - v_\infty) \\ &= -g[v - (D + B/g)] \\ &= g[D + B/g - v] \\ &= g(D - v) + B \quad \blacksquare \end{aligned} \quad (5)$$

Optional Approach 2: Some students may have learned an approach using the integrating factor from a previous course. This is also valid, although not required on the exam.

Arrange (1) to apply integrating factor method:

$$\begin{aligned}\dot{v} &= g \cdot (D - v) + B \\ \dot{v} + gv &= \underbrace{gD + B}_{gv_\infty}\end{aligned}\tag{6}$$

The integrating factor is $\exp(\int_{dt} g) = e^{gt}$; multiply through:

$$\begin{aligned}e^{gt} [\dot{v} + gv] &= e^{gt} gv_\infty \\ e^{gt} \dot{v} + g e^{gt} v &= e^{gt} gv_\infty \\ \frac{d}{dt} [e^{gt} v] &= e^{gt} gv_\infty\end{aligned}\tag{7}$$

Integrate both sides and collect terms:

$$\begin{aligned}\int_{dt} \frac{d}{dt} [e^{gt} v] &= \int_{dt} e^{gt} gv_\infty \\ e^{gt} v(t) &= v_\infty e^{gt} + c \\ v(t) &= v_\infty + c e^{-gt}\end{aligned}\tag{8}$$

Evaluate at $t = 0$ to get constant c :

$$\begin{aligned}v_0 &= v_\infty + c e^{-g \cdot 0} = v_\infty + c \\ c &= v_0 - v_\infty \\ v &= v_\infty + \underbrace{e^{-gt}}_{\alpha(t)} (v_0 - v_\infty) \\ v &= v_\infty + \alpha(t) v_0 - \alpha(t) v_\infty \\ v &= \alpha(t) v_0 + [1 - \alpha(t)] v_\infty\end{aligned}\tag{9}$$

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Q2: Provide the solution to $\tau \dot{v} = c - v$ and evaluate $v(t)$ at time $t = 10$ for $c = 0$, $\tau = 10$, and $v(0) = -70$.

The first part (“provide solution”) is similar to Q1 and we are not asked to do redundant work. If you can intuit the solution immediately, simply write it. Otherwise, any approach valid for Q1 will also get you here. If you’re not 100% confident, you can differentiate to verify.

$$v(t) = c + e^{-t/\tau} [v(0) - c]$$

Evaluate at given values:

$$\begin{aligned} v(10) &= 0 + e^{-10/10} [(-70) - 0] && \text{(plugging in constants' values)} \\ &= -\frac{1}{e} \cdot 70 && \text{(evaluating)} \end{aligned} \tag{10}$$

There’s no need to use a calculator to go any further (a correct numeric answer with incorrect algebra might raise suspicion). Any variation algebraically equivalent to the above will get full marks.

Note: this reminds us that the time constant τ is the time it takes for the distance between $v(0)$ and c to decay to $1/e$ of its original value.

Q3:

You are asked to write computer code to numerically integrate a linear first-order ODE for a neuron's membrane voltage v , being driven by a known time-varying input $u(t)$

$$C\dot{v} = \frac{1}{R}(E - v) + u(t), \quad (11)$$

where C , R , and E are constants.

Q3a: (facts/trivia) Is forward Euler a first or second-order method?

First

Q3b: (facts/trivia) What is the order of the error in forward Euler in terms of the time step Δt ? (What about over a fixed time interval T ?)

$O(\Delta t^2)$ ($O(\Delta t)$)

Q3c: You're told that the signal $u(t)$ varies slowly, and is approximately constant $u(t) \approx u_t$ over a short duration between t and $t + \Delta t$, where $\Delta t = 1$ ms. You implement the following (pseudo)code for *forward exponential Euler* to integrate from $t = 0$ to $t = 1000$ ms based on this assumption:

$$\begin{aligned} \gamma &= ??? \\ v &= v_0 \\ \text{for } t \text{ in } 1 \dots 1000 : \\ &v_\infty = ??? \\ &\Delta_v = \gamma(v_\infty - v_t) \\ &v_{t+1} = v_t + \Delta_v \end{aligned} \quad (12)$$

What expressions belong in the ??? for variables γ and v_∞ ?

Solve assuming constant $u(t) \approx u_t$:

$$\begin{aligned} C\dot{v} &= \frac{1}{R}(E - v) + u_t \\ \underbrace{RC}_{\tau} \dot{v} &= \underbrace{E + Ru_t}_{v_\infty} - v \\ v(t) &= v_\infty + e^{-t/\tau}(v - v_\infty) \end{aligned} \quad (13)$$

Every time-step solve forward by Δt using a new u_t :

$$\begin{aligned} v_{t+1} &= v_\infty + \underbrace{e^{-\Delta t/\tau}}_{\alpha}(v_t - v_\infty) \\ \Delta_v = v_{t+1} - v_t &= v_\infty + \alpha(v_t - v_\infty) - v_t \\ &= \underbrace{(1 - \alpha)}_{\gamma}(v_\infty - v_t) \end{aligned} \quad (14)$$

Full marks for any technically valid response, even if it is not an exact match to the one above.